

Title

ZPL_Operand_Non_Sequential_Extract_FWHM_from_detector

Author

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Brief Description

This macro will extract Detector Viewer cross-section data and calculate the approximate FWHM value.

Overview

This macro will calculate the approximate FWHM of a near-Normal distribution of data. This macro assumes symmetry about the maximum value and assumes a minimal amount of noise between 0 and the maximum. The macro will update the Detector Viewer settings to a cross-section view. Then, the data will be extracted and evaluated. When using the ZPL10 version of the macro, the approximate FWHM will be reported when Data = 0 for the X-direction, and Data = 1 for the Y-Direction. Use the Hx entry to specify the detector of interest.

Plugin Type

ZPL Operand

Language

ZPL

Updates

Date	Version	OpticStudio version	Comment
2019/10/15	1.0	19.4SP2	Creation
2020/10/14	2.0	20.3	Updated macro with the following: • CFG file is programmatically calculated • FWHM index location calculation updated to match Excel LOOKUP function • FWHM index location calculation repaired to check for +/- index value location
2021/10/22	3.0	21.3	Updated macro with the following: • CFG filename search has been updated. The previous method searched for any CFG file. The new method will search for the file-specific CFG file. This is useful if

			the lens file is stored in the same directory as others. • The calculation has been updated. Previously, the calculation assumed perfect symmetry of the array of data. This caused the calculation to fail if the distribution was not centered on the detector. Now, the HWHM is calculated for one side and doubled. • The FWHM calculation is now more robust, finding the exact location of the half max instead of using the closest available location in the array of intensity data.
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Instructions

1. Installation

Copy the macro to the {Zemax}\Macros folder

2. How to use

In OpticStudio, navigate to **Programming...Edit/Run** and open the macro for editing. Update the ray-trace settings on line 36. This is used to populate the Detector Viewer data:

```
32 ! Prepare the files
33 ! Run a ray trace with the appropriate settings
34   clear = NSDD(1, 0, 0, 0)           # Clear all detectors
35   UPDATE ALL
36   NSTR 1, 0, 0, 0, 1, 1, 0, 0       # Run a ray trace
37
```

The ray trace settings are as follows:

Syntax:

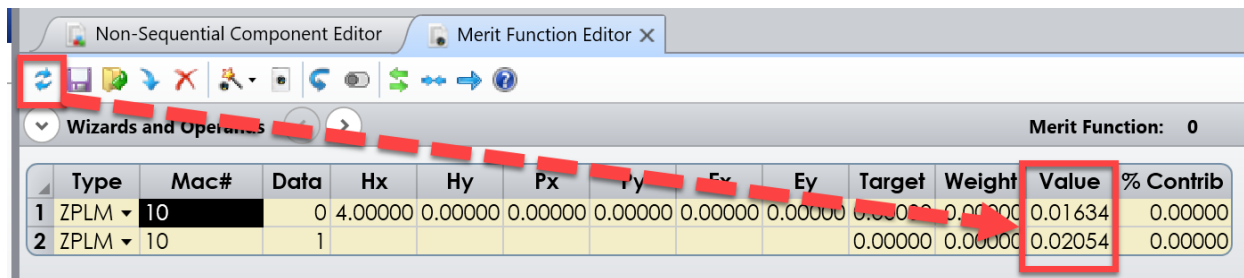
```
NSTR surf, source, split, scatter, usepolar, ignore_errors, random_seed, save, savefilename, filter, zrd_format
```

By default, the ray trace assumes all sources are used. Additionally, the ray trace will Use Polar and Ignore Errors.

Once updated, save the macro as a ZPL##.ZPL. Then you will be able to insert it as a ZPLM operand. In the screenshot provided, the macro is ZPL10.ZPL and is installed into the Merit Function with **Mac#** = 10.

Once inserted into the Merit Function, you can specify the detector of interest by entering a value for **Hx**. When Data = 0, the FWHM in the X-direction will be reported. When Data = 1, the FWHM in the Y-direction will be reported. For more information on ZPL macro operands, see the KBA "[Optimization using a ZPL Macro: the ZPLM operand](#)."

Click the "Update" button in the Merit Function to see the results:



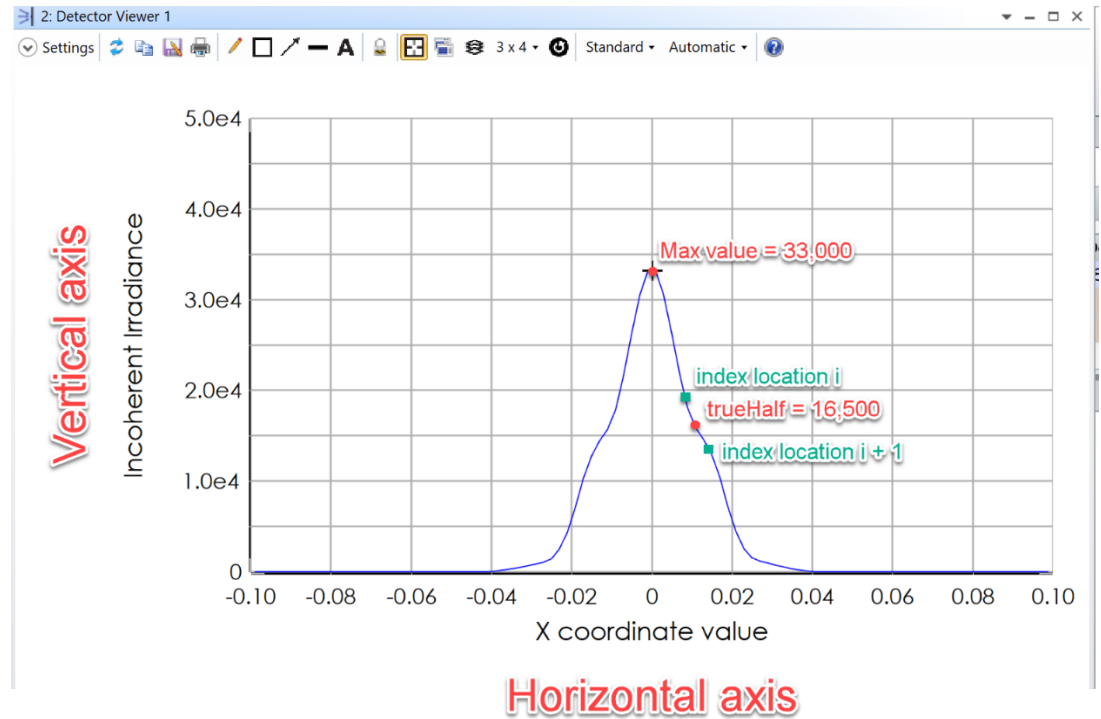
Type	Mac#	Data	Hx	Hy	Px	Py	Fx	Fy	Target	Weight	Value	% Contrib
1 ZPLM	10	0	4.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.01634	0.00000
2 ZPLM	10	1							0.00000	0.00000	0.02054	0.00000

Note: This macro was tested with Detector 4 (slightly displaced) within the Even Asphere.ZOS sample file.

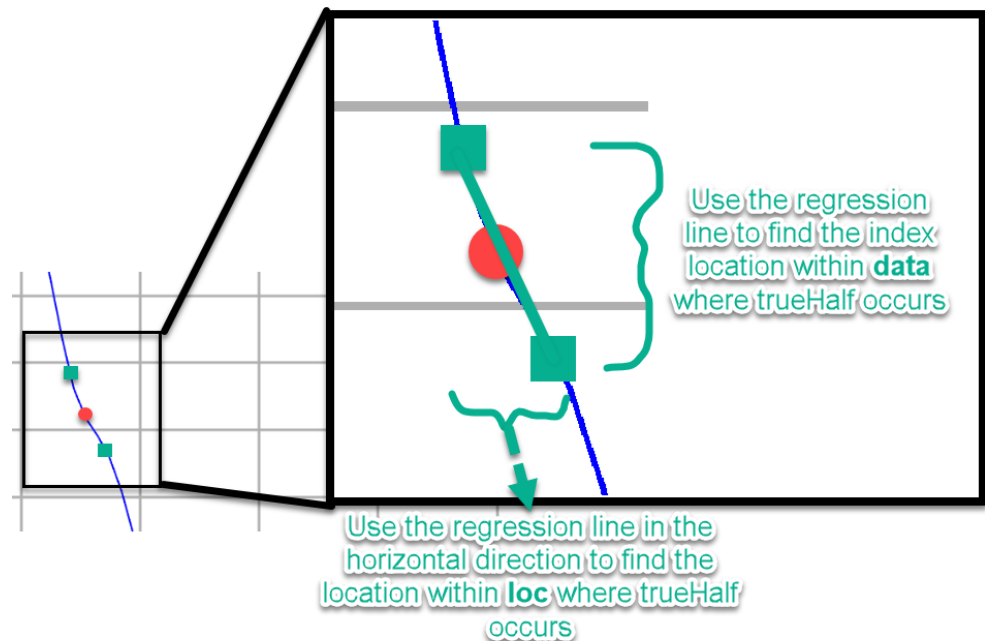
General set of steps:

For each direction, the following set of steps is performed:

1. Store the detector number of interest. This is determined by entering a value under the **Hx** column in the Merit Function. If you are running this macro outside of the Merit Function, then the detector value must be manually entered on line 13.
2. Find the Detector Viewer configuration file. The code will look for the file-specific configuration file. If none is present, it will use DVR.CFG from the Configs folder. DVR.CFG is generated after the first time you save a detector output in any file.
3. Run a Non-Sequential ray trace within the macro using the NSTR command. Note: it may be cleaner to run a ray trace within the Merit Function with the NSTR operand and suppress line 38.
4. Extract the data. We will first update the detector settings to view **Cross Section Row** or **Cross Section Column**. All other Detector Viewer settings will remain in place. The data will be stored in two arrays: **data** and **loc**. The vertical axis data is stored in **data** whereas the horizontal axis information is stored in **loc**.
5. Cycle through the **data** array to find the maximum intensity value. Record the value and its corresponding location from the **loc** array. Divide the maximum value by two and store the information as *trueHalf*.
6. Begin the search for the *trueHalf* value on the right-hand side of the maximum value.
 - a. The search will find the index location where the **data** value is approximately equal to the *trueHalf* value.
 - b. If the **data** value at index location *i* is larger than *trueHalf*, then index location *i* and *i+1* are extracted. If index location *i* is smaller than *trueHalf*, then index location *i* and *i-1* are extracted.



- c. Using the two index locations, we will form a line of best fit for the intensity information provided in the **data** array.
- d. Using the two index locations, we will form a line of best fit for the location information provided in the **loc** array.
- e. Using the two regression lines, find the location of the *trueHalf* on the vertical and horizontal axis of the given view.



7. Report the distance between the location of the max and the positive direction half width along the horizontal axis. Multiply this by 2
8. Report the multiplied value as the FWHM for the X or Y direction.

Assumptions:

1. It is assumed that the detector cross-section view is approximately normal. This means that there is a single defined peak and a low amount of noise. Smoothing the data on the Detector Viewer will work, but only if those settings are **Saved** before running the macro
2. The e^2 calculation has been suppressed in this iteration, but can be added by following the calculations here: "[How to convert FWHM measurements to \$1/\(e^2\)\$ halfwidths.](#)"